@ JeongHo SEO

Time Series Classification for 연도별 trend 반영

- 2025년도에 상장폐지될 확률? ⇒ sliding window 사용. window size = k (tuning parameter, 5 사용.)
- N-k, ..., N-1 recent k개 년도 data → N 년도 상장폐지 확률?
- each sample = (N-k, ..., N-1 recent k개 년도 input feature, N 년도 상장폐지 label)
- dataset = all samples, 기업별 구분 무시 (k개 년도에 따른 label값만 중요)
  - ⇒ NA처리 후 기업별 row 개수 차이 무시.

구분	예측 연도 구간	입력 연도 구간 (k = 5 기준)
Train	2004 ~ 2016	1999 ~ 2015
Validation	2017 ~ 2020	2012 ~ 2019
Test	2021 ~ 2024	2016 ~ 2023

실제 예측 (2025 상장폐지 예측)

- 2020~2024년 재무정보 (최근 5년)를 이용하여 2025년 상장폐지 확률 예측
- 학습된 모델에 해당 윈도우만 inference용으로 입력
- 실제 y-label은 없음 → 예측값 (확률)만 산출

Models

- MLP
- GRU
- TCN
- Ensemble

전처리

1. feature selection
   - ratio 처리
   - 시계열 data: 회계년 변수 필요
   - target feature 관련 변수 포함
     - Symbol(기업 구분용), 상장폐지일자
2. 기업별로 each row마다 inf & NA 처리
   - inf
   - NA가 적은(10% 미만) row 생존
   - NA가 많은(30% 이상) row 제거
   - NA가 애매한(10%~30%) row는 선형 보간법 사용.
3. 기업별로 첫 상장폐지 이후 row 제거 [회생하는 경우 배제]

4. k개 년도 연속 sliding window 생성

- 불연속 window 제거: 연도가 연속적이지 않은 경우에는 해당 sample 제거. (ex. 2000, **2001, 2003**, 2004, 2005)
- each sample마다 y-label 부여.
  - 0: 상장유지, 1: 상장폐지

Input Features Selection @ 홍인택님 추후 수정

- 연도 (회계년) => 결산월은 무시. (12월 결산 대다수, but 일부 3월, 6월 존재.)
- 금융비용/부채
- 금융비용/총비용
- 설비투자효율 = 매출액/유형자산
- 현금흐름/총자본
- 영업이익/총자본
- 유보이익/총자본
- 자기자본비율 = 자기자본/총자산
- 자본금회전율 = 매출액/자본금
- 차입금의존도 = 차입금/총자산
- 총부채회전율 = 매출액/총부채
- 총자본투자효율 = 영업이익/총자산
- 총자본회전율 = 매출액/총자산
- 총순자산수익률 = 당기순이익/자기자본
- 현금성자산비율 = 현금및현금성자산/총자산

In [ ]:
```
! pip install pandas
! pip install torch
! pip install scikit-learn
```

In [214…
```
import pandas as pd
import numpy as np

import random
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import DataLoader, TensorDataset
from sklearn.metrics import accuracy_score
```

1. main_Dataset_mod.csv Preview

In [37]:
```
df = pd.read_csv('main_dataset_mod.csv')

print(df.shape)
print(df.head())
print(df.columns.to_list())
```

```
C:\Users\Public\Documents\ESTsoft\CreatorTemp\ipykernel_9352\1713723279.py:1: Dty
peWarning: Columns (260,261) have mixed types. Specify dtype option on import or
set low_memory=False.
  df = pd.read_csv('main_dataset_mod.csv')
```

```
(163046, 265)
    Symbol  Name   결산월   회계년      주기       총자산(천원)   총자산(평균)(천
원)  유동자산(천원)  \
0  A000010  조흥은행  12.0  1999  Annual  4.693918e+10  4.674953e+10       NaN
1  A000010  조흥은행  12.0  2000  Annual  5.113869e+10  4.903894e+10       NaN
2  A000010  조흥은행  12.0  2001  Annual  5.661012e+10  5.387440e+10       NaN
3  A000010  조흥은행  12.0  2002  Annual  6.727043e+10  6.194027e+10       NaN
4  A000010  조흥은행  12.0  2003  Annual  6.060160e+10  6.393601e+10       NaN

   당좌자산(천원)  현금및현금성자산(천원)  ...  현금흐름(연율화)(천원)  현금흐름
(직전4분기)(천원)  현금흐름(보통)(천원)  \
0       NaN  1.210399e+09  ...   -746322000.0             NaN  -746322000.0
1       NaN  9.738450e+08  ...     99359000.0             NaN    99359000.0
2       NaN  1.346084e+09  ...    710634000.0             NaN   710634000.0
3       NaN  1.174976e+09  ...   -395695000.0             NaN  -395695000.0
4       NaN  9.181470e+08  ...   -750930000.0             NaN  -750930000.0

   현금흐름(보통,연율화)(천원)  현금흐름(보통,직전4분기)(천원)                상장
폐지일자  \
0      -746322000.0                NaN  2004-07-02T00:00:00Z
1        99359000.0                NaN  2004-07-02T00:00:00Z
2       710634000.0                NaN  2004-07-02T00:00:00Z
3      -395695000.0                NaN  2004-07-02T00:00:00Z
4      -750930000.0                NaN  2004-07-02T00:00:00Z

                      폐지사유  파산여부  부도여부  Bad
0  지주회사의 완전자회사화(지주회사 신규상장)     0     0    0
1  지주회사의 완전자회사화(지주회사 신규상장)     0     0    0
2  지주회사의 완전자회사화(지주회사 신규상장)     0     0    0
3  지주회사의 완전자회사화(지주회사 신규상장)     0     0    0
4  지주회사의 완전자회사화(지주회사 신규상장)     0     0    0

[5 rows x 265 columns]
['Symbol', 'Name', '결산월', '회계년', '주기', '총자산(천원)', '총자산(평균)(천
원)', '유동자산(천원)', '당좌자산(천원)', '현금및현금성자산(천원)', '단기금융자산
(천원)', '단기금융상품(천원)', '단기유가증권(천원)', '단기대여금(천원)', '매출채권
및기타채권(천원)', '매출채권(천원)', '단기미청구공사(천원)', '재고자산(천원)', '당
기법인세자산(천원)', '매각예정비유동자산(천원)', '비유동자산(천원)', '장기금융자산
(천원)', '장기금융상품(천원)', '매도가능금융자산(천원)', '만기보유금융자산(천원)',
'장기대여금(천원)', '투자부동산(천원)', '관계기업등지분관련투자자산(천원)', '종속
기업투자(천원)', '관계기업투자등(천원)', '유형자산(천원)', '토지(천원)', '설비자산
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원)', '회원권및제이용권(천원)', '개발비(천원)', '산업재산권(천원)', '비유동생물자
산(천원)', '장기매출채권및기타채권(천원)', '이연법인세자산(천원)', '기타금융업자산
(천원)', '*단기유동성(천원)', '*총금융자산(천원)', '*총유가증권(천원)', '부채및자
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채(천원)', '*총금융리스부채(천원)', '총자본(천원)', '총자본(평균)(천원)', '지배주
주지분(천원)', '지배주주지분(평균)(천원)', '지배주주순자산(자사주차감전)(천원)',
'자본금(천원)', '보통주자본금(천원)', '우선주자본금(천원)', '자본금(평균)(천원)',
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익(천원)', '기타자본잉여금(천원)', '기타자본구성요소(천원)', '자기주식(천원)', '자
기주식처분손실(천원)', '이익잉여금(천원)', '기타포괄손익누계액(천원)', '금융자산평
```

가손익(천원)', '매도가능금융자산평가손익(천원)', '관계기업등기타포괄손익(천원)', '환산관련외환차이적립금(환율변동차이)(천원)', '재평가잉여금(천원)', '비지배주주지분(천원)', '*유보액(천원)', '*유보액(3년평균)(천원)', '매출액(천원)', '매출액(3년평균)(천원)', '매출액(연율화)(천원)', '매출액(직전4분기)(천원)', '매출원가(천원)', '매출총이익(천원)', '판매비와관리비(천원)', '인건비(천원)', '급여(천원)', '퇴직급여(천원)', '복리후생비(천원)', '유무형자산상각비(천원)...121', '유형자산감가상각비(천원)...122', '무형자산상각비(천원)...123', '연구개발비(천원)', '광고선전비(천원)', '판매비(천원)', '대손상각비(판관비)(천원)', '관리비(천원)', '임차료(천원)', '세금과공과(천원)', '영업이익(천원)', '영업이익(3년평균)(천원)', '영업이익(연율화)(천원)', '영업이익(직전4분기)(천원)', '*조정영업이익(천원)', '*조정영업이익(직전4분기)(천원)', '비영업손익(천원)', '금융손익(비영업)(천원)', '관계기업투자등관련손익(비영업)(천원)', '기타손익(비영업)(천원)', '외환관련손익(비영업)(천원)', '자산재평가손익(천원)', '비영업수익(천원)', '금융수익(비영업)(천원)', '관계기업투자등관련이익(비영업)(천원)', '기타수익(비영업)(천원)', '이자수익(비영업)(천원)', '배당금수익(비영업)(천원)', '외환이익(비영업)(천원)', '외환차익(비영업)(천원)', '외환환산이익(비영업)(천원)', '유가증권관련이익(비영업)(천원)', '비영업비용(천원)', '금융원가(비영업)(천원)', '관계기업투자등관련손실(비영업)(천원)', '기타비용(비영업)(천원)', '이자비용(비영업)(천원)', '*순이자비용(비영업)(천원)', '외환손실(비영업)(천원)', '외환차손(비영업)(천원)', '외화환산손실(비영업)(천원)', '유가증권관련손실(비영업)(천원)', '세전계속사업이익(천원)', '세전계속사업이익(3년평균)(천원)', '세전계속사업이익(연율화)(천원)', '세전계속사업이익(직전4분기)(천원)', '계속사업법인세비용(천원)', '계속사업이익(천원)', '계속사업이익(직전4분기)(천원)', '중단사업이익(천원)', '중단사업이익(직전4분기)(천원)', '당기순이익(천원)', '당기순이익(3년평균)(천원)', '당기순이익(연율화)(천원)', '당기순이익(직전4분기)(천원)', '지배주주순이익(천원)', '지배주주순이익(3년평균)(천원)', '지배주주순이익(연율화)(천원)', '지배주주순이익(직전4분기)(천원)', '비지배주주순이익(천원)', '기타포괄손익(천원)', '총포괄이익(천원)', '지배주주총포괄이익(천원)', '비지배주주총포괄이익(천원)', '순영업이익(천원)', '영업수익(천원)', '영업수익(3년평균)(천원)', '영업수익(연율화)(천원)', '영업수익(직전4분기)(천원)', '영업비용(천원)', '영업활동으로인한현금흐름(천원)', '영업활동으로인한현금흐름(직전4분기)(천원)', '영업활동으로인한현금흐름(평균)(천원)', '당기순손익(법인세차감전순손익)(천원)', '현금유출입이없는비용(수익)(천원)', '현금유출이없는비용의가산(천원)', '유무형자산상각비(천원)...197', '유형자산감가상각비(천원)...198', '무형자산상각비(천원)...199', '이자비용(천원)', '외환손실(천원)', '유가증권관련손실(천원)', '관계회사등투자손실(천원)', '법인세비용(천원)', '외화환산순손실(이익)(천원)', '현금유입이없는수익의차감(천원)', '이자수익(천원)', '외환이익(천원)', '유가증권관련이익(천원)', '관계회사등투자이익(천원)', '영업활동으로인한자산및부채의변동(천원)', '영업활동으로인한자산의감소(천원)', '매출채권및기타채권의감소(증가)(천원)', '재고자산의감소(증가)(천원)', '영업활동으로인한부채의증가(천원)', '매입채무및기타채무의증가(감소)(천원)', '기타영업자산및부채의변동(천원)', '투자활동으로인한현금흐름(천원)', '투자활동으로인한현금유입액(천원)', '투자활동으로인한현금유출액(천원)', '유형자산처분(취득)(천원)', '무형자산감소(증가)(천원)', '*유형자산순취득액(천원)', '*무형자산순취득액(천원)', '장단기금융자산의감소(증가)(천원)', '단기금융자산의감소(증가)(천원)', '장기금융자산의감소(증가)(천원)', '기타투자활동으로인한현금흐름(천원)', '재무활동으로인한현금흐름(천원)', '재무활동으로인한현금유입액(천원)', '장단기금융부채의증가(천원)', '자본의증가(천원)', '재무활동으로인한현금유출액(천원)', '장단기금융부채의감소(천원)', '자본의감소(천원)', '장단기금융부채의증가(감소)(천원)', '단기금융부채의증가(감소)(천원)', '장기금융부채의증가(감소)(천원)', '자본의증가(감소)(천원)', '기타재무활동으로인한현금흐름(천원)', '이자수취(영업,투자,재무)(천원)', '이자지급(영업,투자,재무)(천원)', '배당금수취(영업,투자,재무)(천원)', '배당금지급(영업,투자,재무)(천원)', '법인세환급(영업,투자,재무)(천원)', '법인세납부(영업,투자,재무)(천원)', '기타현금흐름(천원)', '영업,투자,재무활동기타현금흐름(천원)', '연결범위변동으로인한현금의증가(천원)', '환율변동효과(천원)', '합병분할(영업양수도)으로인한증가(감소)(천원)', '현금의순증가(천원)', '기초현금(천원)', '기말현금(천원)', '현금흐름(천원)', '현금흐름(연율화)(천원)', '현금흐름(직전4분기)(천원)', '현금흐름(보통)(천원)', '현금흐름(보통,연율화)(천원)', '현금흐름(보통,직전4분기)(천원)', '상장폐지일자', '폐지사유', '파산여부', '부도여부', 'Bad']

2. 전처리

- Feature Selection

In [38]:
```
df["연도"] = df["회계년"].astype(int)
df["상장폐지년도"] = pd.to_datetime(df["상장폐지일자"], errors="coerce").dt.year

# 파생변수 생성
df['금융비용/부채'] = df['이자비용(비영업)(천원)'] / df['총부채(천원)']

total_cost = df['매출원가(천원)'] + df['판매비와관리비(천원)']
df['금융비용/총비용'] = df['이자비용(비영업)(천원)'] / total_cost

df['설비투자효율'] = df['매출액(천원)'] / df['유형자산(천원)']

df['현금흐름/총자본'] = df['영업활동으로인한현금흐름(천원)'] / df['총자본(천원)']

df['영업이익/총자본'] = df['영업이익(천원)'] / df['총자본(천원)']

df['유보이익/총자본'] = df['이익잉여금(천원)'] / df['총자본(천원)']

df['자기자본비율'] = df['총자본(천원)'] / df['총자산(천원)']

df['자본금회전율'] = df['매출액(천원)'] / df['자본금(천원)']

total_loans = df['단기차입금(천원)'] + df['장기차입금(천원)']
df['차입금의존도'] = total_loans / df['총자산(천원)']

df['총부채회전율'] = df['매출액(천원)'] / df['총부채(천원)']

df['총자본투자효율'] = df['영업이익(천원)'] / df['총자산(천원)']

df['총자본회전율'] = df['매출액(천원)'] / df['총자산(천원)']

df['총순자산수익률'] = df['당기순이익(천원)'] / df['총자본(천원)']

df['현금성자산비율'] = df['현금및현금성자산(천원)'] / df['총자산(천원)']


selected_columns = [
        "Symbol", "연도", "상장폐지년도",
        '금융비용/부채', '금융비용/총비용', '설비투자효율', '현금흐름/총자본',
        '영업이익/총자본', '유보이익/총자본', '자기자본비율', '자본금회전율',
        '차입금의존도', '총부채회전율', '총자본투자효율', '총자본회전율',
        '총순자산수익률', '현금성자산비율'
        ]

df = df[selected_columns].copy()
```

- inf & NA 처리

In [ ]:
```
def process_missing_and_inf(group):
    group = group.replace([np.inf, -np.inf], np.nan) # inf/-inf -> NaN
    row_na_ratio = group.isna().mean(axis=1)

    # NA 비율 기
    cond_low = row_na_ratio < 0.10 # 유지
    cond_mid = (row_na_ratio >= 0.10) & (row_na_ratio < 0.30) # 선형보간법
    cond_high = row_na_ratio >= 0.30 # 제거
```

```
    # 제거 + 보간
    group = group[~cond_high] # 제거
    group.loc[cond_mid] = group.loc[cond_mid].interpolate(method='linear', limit

    return group.dropna() # 남은 NaN 제거

df = df.groupby('Symbol', group_keys=False).apply(process_missing_and_inf)
```

```
C:\Users\Public\Documents\ESTsoft\CreatorTemp\ipykernel_9352\488844392.py:12: Fut
ureWarning: DataFrame.interpolate with object dtype is deprecated and will raise
in a future version. Call obj.infer_objects(copy=False) before interpolating inst
ead.
  group.loc[cond_mid] = group.loc[cond_mid].interpolate(method='linear', limit_di
rection='both')
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C:\Users\Public\Documents\ESTsoft\CreatorTemp\ipykernel_9352\488844392.py:16: Dep
recationWarning: DataFrameGroupBy.apply operated on the grouping columns. This be
havior is deprecated, and in a future version of pandas the grouping columns will
be excluded from the operation. Either pass `include_groups=False` to exclude the
groupings or explicitly select the grouping columns after groupby to silence this
warning.
  df = df.groupby('Symbol', group_keys=False).apply(process_missing_and_inf)
```

- 상장폐지 이후 row 제거

In [70]:
```python
def remove_after_delisting(group):
    if group['상장폐지년도'].notna().any():
        first_delist_year = group['상장폐지년도'].dropna().iloc[0]
        return group[group['연도'] <= first_delist_year]
    return group

df = df.groupby('Symbol', group_keys=False).apply(remove_after_delisting)
```

```
C:\Users\Public\Documents\ESTsoft\CreatorTemp\ipykernel_9352\2732059550.py:7: Dep
recationWarning: DataFrameGroupBy.apply operated on the grouping columns. This be
havior is deprecated, and in a future version of pandas the grouping columns will
be excluded from the operation. Either pass `include_groups=False` to exclude the
groupings or explicitly select the grouping columns after groupby to silence this
warning.
  df = df.groupby('Symbol', group_keys=False).apply(remove_after_delisting)
```

- k개 년도 sliding window 생성

In [88]:
```python
def generate_sliding_windows(df, k):
    samples = []
    grouped = df.groupby('Symbol')

    for symbol, group in grouped:
        group = group.sort_values('연도')
        years = group['연도'].values
        features = group.drop(columns=['Symbol', '상장폐지년도']).values

        for i in range(len(group) - k):
            window_years = years[i:i + k + 1]
            if not np.all(np.diff(window_years) == 1):  # 연속 연도 아닌 경우 제
                continue

            X_window = features[i:i + k] # k개 년도의 연속된 feature
            prediction_year = years[i + k] # 예측할 연도
            y_delist_year = group.iloc[i + k]['상장폐지년도'] # 실제 상장폐지년도
```

```
            # 상장폐지년도 = 예측년도 or 예측년도 + 1 ... 재무재표가 너무 안좋아서
            if pd.notna(y_delist_year) and (y_delist_year == prediction_year or
                y_label = 1
            else:
                y_label = 0

            samples.append((X_window, y_label, prediction_year))

    return samples

k = 5 # tuning parameter
samples = generate_sliding_windows(df, k) # X_window, y_label, prediction_year
```

- y-label distriution

In [101…
```
def check_label_distribution(dataset, name="Dataset"):
    # (X, y) 또는 (X, y, year) 모두 처리 가능
    labels = [y for item in dataset for i, y in enumerate(item) if i == 1]
    unique, counts = np.unique(labels, return_counts=True)

    print(f"\n{name} 샘플 수: {len(labels)}")
    for u, c in zip(unique, counts):
        print(f"- Label {u}: {c}개 ({c / len(labels):.2%})")


check_label_distribution(samples)
```

```
Dataset 샘플 수: 5723
- Label 0: 5004개 (87.44%)
- Label 1: 719개 (12.56%)
```

- sample: X_window, y_label, prediction_year

In [100…
```
print(f"1st sample example: \n{samples[0]}\n")

print(f"1st sample input: {samples[0][0]}")
print(f"1st sample label: {samples[0][1]}")
if samples[0][1] == 1:
    print(f"1st sample 상장폐지년도: {samples[0][2]}")
else:
    print(f"1st sample: 상장폐지 X")
print(f"1st sample 입력연도: {samples[0][2]-5}-{samples[0][2]-1}")
print(f"1st sample 예측연도: {samples[0][2]}")
```

```
1st sample example:
(array([[1.99900000e+03, 5.83748687e-02, 5.88561937e-02, 1.88183020e+00,
        1.06919958e-01, 3.40049968e-02, 9.69881651e-02, 4.20238959e-01,
        8.34104779e+00, 1.21920478e-01, 1.01647049e+00, 1.42902244e-02,
        5.89309989e-01, 1.07458378e-02, 3.23672040e-02],
       [2.00000000e+03, 7.93253225e-02, 8.11612473e-02, 1.84995904e+00,
        3.68445013e-01, 1.22090766e-01, 1.04085401e-01, 4.47388931e-01,
        7.83414463e+00, 9.22771799e-02, 1.07622287e+00, 5.46220571e-02,
        5.94732672e-01, 1.53163684e-02, 0.00000000e+00],
       [2.00100000e+03, 5.33285900e-02, 4.64396303e-02, 1.99845195e+00,
        2.04000353e-01, 6.55651292e-02, 1.27612911e-01, 4.66614382e-01,
        8.35753038e+00, 1.09464952e-01, 1.20569970e+00, 3.05936322e-02,
        6.43102878e-01, 3.93582686e-02, 2.75747851e-02],
       [2.00200000e+03, 4.60051371e-02, 3.67444065e-02, 1.86833933e+00,
        2.01674783e-01, 8.65159861e-02, 1.48542906e-01, 5.23006820e-01,
        7.61531671e+00, 7.89129379e-02, 1.34689287e+00, 4.52484508e-02,
        6.42458714e-01, 3.77884368e-02, 6.64935681e-02],
       [2.00300000e+03, 2.59236508e-02, 1.73660151e-02, 2.06906850e+00,
        9.71110185e-02, 5.06041465e-02, 1.81525646e-01, 5.54236984e-01,
        8.16327462e+00, 8.61743137e-02, 1.55569901e+00, 2.80466895e-02,
        6.93473082e-01, 4.27007376e-02, 6.71647280e-02]]), 0, np.int64(2004))

1st sample input: [[1.99900000e+03 5.83748687e-02 5.88561937e-02 1.88183020e+00
  1.06919958e-01 3.40049968e-02 9.69881651e-02 4.20238959e-01
  8.34104779e+00 1.21920478e-01 1.01647049e+00 1.42902244e-02
  5.89309989e-01 1.07458378e-02 3.23672040e-02]
 [2.00000000e+03 7.93253225e-02 8.11612473e-02 1.84995904e+00
  3.68445013e-01 1.22090766e-01 1.04085401e-01 4.47388931e-01
  7.83414463e+00 9.22771799e-02 1.07622287e+00 5.46220571e-02
  5.94732672e-01 1.53163684e-02 0.00000000e+00]
 [2.00100000e+03 5.33285900e-02 4.64396303e-02 1.99845195e+00
  2.04000353e-01 6.55651292e-02 1.27612911e-01 4.66614382e-01
  8.35753038e+00 1.09464952e-01 1.20569970e+00 3.05936322e-02
  6.43102878e-01 3.93582686e-02 2.75747851e-02]
 [2.00200000e+03 4.60051371e-02 3.67444065e-02 1.86833933e+00
  2.01674783e-01 8.65159861e-02 1.48542906e-01 5.23006820e-01
  7.61531671e+00 7.89129379e-02 1.34689287e+00 4.52484508e-02
  6.42458714e-01 3.77884368e-02 6.64935681e-02]
 [2.00300000e+03 2.59236508e-02 1.73660151e-02 2.06906850e+00
  9.71110185e-02 5.06041465e-02 1.81525646e-01 5.54236984e-01
  8.16327462e+00 8.61743137e-02 1.55569901e+00 2.80466895e-02
  6.93473082e-01 4.27007376e-02 6.71647280e-02]]
1st sample label: 0
1st sample: 상장폐지 X
1st sample 입력연도: 1999-2003
1st sample 예측연도: 2004
```

- dataset 분할

In [102…

```python
train_set, val_set, test_set = [], [], []

for X_window, y_label, prediction_year in samples:
    if 2004 <= prediction_year <= 2016:
        train_set.append((X_window, y_label))
    elif 2017 <= prediction_year <= 2020:
        val_set.append((X_window, y_label))
    elif 2021 <= prediction_year <= 2024:
        test_set.append((X_window, y_label))
```

```
# 라벨 없음, 2025년도 상장폐지 예측
def extract_inference_windows(df, k=5, target_year=2025):
    infer_set = []
    grouped = df.groupby('Symbol')

    for _, group in grouped:
        group = group.sort_values('연도')
        years = group['연도'].values
        features = group.drop(columns=['Symbol', '상장폐지년도']).values

        for i in range(len(group) - k + 1):
            window_years = years[i:i+k]
            if not np.all(np.diff(window_years) == 1):
                continue
            if window_years[-1] == target_year - 1:  # 2020~2024 입력 → 2025 예측
                x_window = features[i:i+k]
                infer_set.append(x_window)
                break

    return np.stack(infer_set) if infer_set else None

X_2025 = extract_inference_windows(df, k=5, target_year=2025)
```

- Case sampling: y label ratio 1:1 ⇒ only for train dataset

In [103…
```
check_label_distribution(train_set, "Train")
check_label_distribution(val_set, "Validation")
check_label_distribution(test_set, "Test")
```

```
Train 샘플 수: 4724
- Label 0: 4241개 (89.78%)
- Label 1: 483개 (10.22%)

Validation 샘플 수: 752
- Label 0: 623개 (82.85%)
- Label 1: 129개 (17.15%)

Test 샘플 수: 247
- Label 0: 140개 (56.68%)
- Label 1: 107개 (43.32%)
```

Test dataset의 상장폐지비율이 43.32%로 큰 것으로 보아, covid-19의 영향이 강세임을 추측할 수 있다.

- train set sample #가 적으므로, weight 방식을 취한다. (CrossEntropyLoss에 weight 인자를 줘서, y=1에 더 높은 손실을 부여)

3. Modeling

- MLP

In [105…
```
class MLPModel(nn.Module):
    def __init__(self, input_dim):
        super().__init__()
        self.net = nn.Sequential(
```

```
            nn.Linear(input_dim, 128),
            nn.ReLU(),
            nn.Dropout(0.3),
            nn.Linear(128, 64),
            nn.ReLU(),
            nn.Linear(64, 2)
        )

    def forward(self, x):
        return self.net(x.view(x.size(0), -1))
```

- GRU

In [106…

```
class GRUModel(nn.Module):
    def __init__(self, input_size, hidden_size=64):
        super().__init__()
        self.gru = nn.GRU(input_size, hidden_size, batch_first=True)
        self.fc = nn.Linear(hidden_size, 2)

    def forward(self, x):
        out, _ = self.gru(x)
        return self.fc(out[:, -1])  # 마지막 시점만
```

- TCN

In [107…

```
class TCNBlock(nn.Module):
    def __init__(self, in_channels, out_channels, kernel_size=2, dilation=1):
        super().__init__()
        padding = (kernel_size - 1) * dilation
        self.conv = nn.Conv1d(in_channels, out_channels, kernel_size,
                              padding=padding, dilation=dilation)
        self.relu = nn.ReLU()
        self.downsample = nn.Conv1d(in_channels, out_channels, 1) if in_channels

    def forward(self, x):
        out = self.conv(x)
        out = out[:, :, :-self.conv.padding[0]]  # causal trim
        return self.relu(out + self.downsample(x))


class TCNModel(nn.Module):
    def __init__(self, input_channels, seq_len):
        super().__init__()
        self.tcn = nn.Sequential(
            TCNBlock(input_channels, 64, dilation=1),
            TCNBlock(64, 64, dilation=2),
        )
        self.fc = nn.Linear(64 * seq_len, 2)

    def forward(self, x):
        x = x.transpose(1, 2)  # (B, C, T)
        x = self.tcn(x)
        return self.fc(x.view(x.size(0), -1))
```

- training & evaluation
- train set에만 가중치 부여

- epoch이 진행될 때마다 accuracy가 항상 증가하지 않고 급락하는 경우가 빈번해서, 0.8 초과 시 stop하게 만들었습니다

In [177…
```
# ✅ 재현성을 위한 시드 고정
SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
torch.cuda.manual_seed_all(SEED)
torch.backends.cudnn.deterministic = True
torch.backends.cudnn.benchmark = False
```

In [178…
```
# ✅ 클래스 가중치 자동 계산 함수
def compute_class_weights(dataset):
    labels = [y for item in dataset for i, y in enumerate(item) if i == 1]
    class_counts = np.bincount(labels, minlength=2)
    weight = class_counts[0] / class_counts.astype(np.float32)
    return torch.tensor(weight, dtype=torch.float32)
```

In [179…
```
# ✅ 학습 + 평가 함수 with Threshold-only 조기 종료

def train_and_evaluate(model, X_train, y_train, X_val, y_val, X_test, y_test, na
    device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
    model.to(device)

    # 클래스 가중치 자동 계산 (train_set에서 받아올 경우)
    if train_set is not None:
        weight = compute_class_weights(train_set).to(device)
    else:
        class_counts = torch.bincount(y_train, minlength=2)
        weight = class_counts[0] / class_counts.float().clamp(min=1)

    criterion_train = nn.CrossEntropyLoss(weight=weight)
    optimizer = torch.optim.Adam(model.parameters(), lr=1e-3)

    train_loader = DataLoader(TensorDataset(X_train, y_train), batch_size=64, sh
    val_loader = DataLoader(TensorDataset(X_val, y_val), batch_size=64)
    test_loader = DataLoader(TensorDataset(X_test, y_test), batch_size=64)

    best_val_acc = 0
    best_model_state = None
    best_epoch = 0

    for epoch in range(1, 101):
        model.train()
        for xb, yb in train_loader:
            xb, yb = xb.to(device), yb.to(device)
            optimizer.zero_grad()
            loss = criterion_train(model(xb), yb)
            loss.backward()
            optimizer.step()

        model.eval()
        val_preds, val_true = [], []
        with torch.no_grad():
            for xb, yb in val_loader:
                out = model(xb.to(device))
                val_preds.extend(out.argmax(dim=1).cpu())
```

```
                val_true.extend(yb)

        acc = accuracy_score(val_true, val_preds)
        print(f"[{name}] Epoch {epoch:02d} - Validation Accuracy: {acc:.4f}")

        if acc > best_val_acc:
            best_val_acc = acc
            best_model_state = model.state_dict()
            best_epoch = epoch

        if best_val_acc >= 0.80:
            print(f"[{name}] Early stopping: Validation Accuracy {best_val_acc:.
            break

    if best_model_state is not None:
        model.load_state_dict(best_model_state)

    model.eval()
    test_preds, test_true = [], []
    with torch.no_grad():
        for xb, yb in test_loader:
            out = model(xb.to(device))
            test_preds.extend(out.argmax(dim=1).cpu())
            test_true.extend(yb)

    test_acc = accuracy_score(test_true, test_preds)
    print(f"\n✅ [{name}] Best Validation Accuracy: {best_val_acc:.4f}")
    print(f"✅ [{name}] Best Epoch: {best_epoch}")
    print(f"✅ [{name}] Test Accuracy on the Best Epoch: {test_acc:.4f}\n")
```

- input data, label 가공

In [151…

```
def unpack(dataset):
    X = np.stack([x for x, _ in dataset])
    y = np.array([y for _, y in dataset])
    return X, y

X_train, y_train = unpack(train_set)
X_val, y_val     = unpack(val_set)
X_test, y_test   = unpack(test_set)
```

In [201…

```
# MLP용: (N, T × D) flatten
X_train_mlp = torch.tensor(X_train.reshape(X_train.shape[0], -1), dtype=torch.fl
X_val_mlp   = torch.tensor(X_val.reshape(X_val.shape[0], -1), dtype=torch.float3
X_test_mlp  = torch.tensor(X_test.reshape(X_test.shape[0], -1), dtype=torch.floa

# GRU/TCN용: (N, T, D) 그대로 유지
X_train_seq = torch.tensor(X_train, dtype=torch.float32)
X_val_seq   = torch.tensor(X_val, dtype=torch.float32)
X_test_seq  = torch.tensor(X_test, dtype=torch.float32)

# y-label 공통
y_train_t = torch.tensor(y_train, dtype=torch.long)
y_val_t   = torch.tensor(y_val, dtype=torch.long)
y_test_t  = torch.tensor(y_test, dtype=torch.long)
```

- Model training & Evaluation

In [202…
```
mlp_model = MLPModel(input_dim=X_train_mlp.shape[1])
train_and_evaluate(
    mlp_model,
    X_train_mlp, y_train_t,
    X_val_mlp, y_val_t,
    X_test_mlp, y_test_t,
    name="MLP",
    train_set=train_set
)
```

```
[MLP] Epoch 01 - Validation Accuracy: 0.8285
[MLP] Early stopping: Validation Accuracy 0.8285 reached threshold 0.80

✅ [MLP] Best Validation Accuracy: 0.8285
✅ [MLP] Best Epoch: 1
✅ [MLP] Test Accuracy on the Best Epoch: 0.5668
```

In [203…
```
gru_model = GRUModel(input_size=X_train_seq.shape[2])
train_and_evaluate(
    gru_model,
    X_train_seq, y_train_t,
    X_val_seq, y_val_t,
    X_test_seq, y_test_t,
    name="GRU",
    train_set=train_set
)
```

```
[GRU] Epoch 01 - Validation Accuracy: 0.8298
[GRU] Early stopping: Validation Accuracy 0.8298 reached threshold 0.80

✅ [GRU] Best Validation Accuracy: 0.8298
✅ [GRU] Best Epoch: 1
✅ [GRU] Test Accuracy on the Best Epoch: 0.5668
```

In [204…
```
tcn_model = TCNModel(input_channels=X_train_seq.shape[2], seq_len=X_train_seq.sh
train_and_evaluate(
    tcn_model,
    X_train_seq, y_train_t,
    X_val_seq, y_val_t,
    X_test_seq, y_test_t,
    name="TCN",
    train_set=train_set
)
```

```
[TCN] Epoch 01 - Validation Accuracy: 0.2221
[TCN] Epoch 02 - Validation Accuracy: 0.3604
[TCN] Epoch 03 - Validation Accuracy: 0.7819
[TCN] Epoch 04 - Validation Accuracy: 0.8045
[TCN] Early stopping: Validation Accuracy 0.8045 reached threshold 0.80

✅ [TCN] Best Validation Accuracy: 0.8045
✅ [TCN] Best Epoch: 4
✅ [TCN] Test Accuracy on the Best Epoch: 0.5749
```

- ensemble

In [205…
```python
def predict_prob(model, X, device='cuda' if torch.cuda.is_available() else 'cpu'
    model.eval()
    model.to(device)

    # numpy array일 경우 tensor로 변환
    if isinstance(X, np.ndarray):
        X = torch.tensor(X, dtype=torch.float32)

    # 예측 확률 계산 (batch 처리)
    probs = []
    with torch.no_grad():
        for i in range(0, len(X), 64):  # 배치 사이즈 64
            xb = X[i:i+64]
            if not isinstance(xb, torch.Tensor):
                xb = torch.tensor(xb, dtype=torch.float32)
            xb = xb.to(device)
            out = model(xb)
            prob = torch.softmax(out, dim=1)[:, 1]  # 클래스 1의 확률만
            probs.append(prob.cpu())
    return torch.cat(probs)
```

In [206…
```python
# 모델 리스트
models = [mlp_model, gru_model, tcn_model]

# 각 모델에 맞는 입력 (주의: 입력 형태가 모델 구조와 맞아야 함)
val_inputs = [X_val_mlp, X_val_seq, X_val_seq]
test_inputs = [X_test_mlp, X_test_seq, X_test_seq]

# Validation 예측 확률
val_probs = [predict_prob(model, X) for model, X in zip(models, val_inputs)]
val_probs_avg = torch.stack(val_probs).mean(dim=0)
val_preds_ensemble = (val_probs_avg >= 0.5).long()
val_acc_ensemble = accuracy_score(y_val_t.numpy(), val_preds_ensemble.numpy())

# Test 예측 확률
test_probs = [predict_prob(model, X) for model, X in zip(models, test_inputs)]
test_probs_avg = torch.stack(test_probs).mean(dim=0)
test_preds_ensemble = (test_probs_avg >= 0.5).long()
test_acc_ensemble = accuracy_score(y_test_t.numpy(), test_preds_ensemble.numpy()

print(f"✅ [Ensemble] Validation Accuracy: {val_acc_ensemble:.4f}")
print(f"✅ [Ensemble] Test Accuracy: {test_acc_ensemble:.4f}")
```

```
✅ [Ensemble] Validation Accuracy: 0.8285
✅ [Ensemble] Test Accuracy: 0.5668
```

4. 2025년 상장폐지 예측

In [207…
```python
# MLP 용: (N, T×D)로 flatten
X_2025_mlp = torch.tensor(X_2025.reshape(X_2025.shape[0], -1), dtype=torch.float

# GRU / TCN 용: (N, T, D) 그대로 사용
X_2025_seq = torch.tensor(X_2025, dtype=torch.float32)
```

In [208…

```python
# 각 모델에 맞는 입력
X_2025_inputs = [X_2025_mlp, X_2025_seq, X_2025_seq]

# 예측 확률 계산
prob_2025_list = [predict_prob(m, x) for m, x in zip(models, X_2025_inputs)]
prob_2025_avg = torch.stack(prob_2025_list).mean(dim=0)
pred_2025_final = (prob_2025_avg >= 0.5).long()

# DataFrame으로 결과 정리
df_pred_2025 = pd.DataFrame({
    "MLP_Prob": prob_2025_list[0].numpy(),
    "GRU_Prob": prob_2025_list[1].numpy(),
    "TCN_Prob": prob_2025_list[2].numpy(),
    "Ensemble_Prob": prob_2025_avg.numpy(),
    "Ensemble_Pred": pred_2025_final.numpy()
})
```

In [211…

```python
print(df_pred_2025.head(10))
print("총 샘플 수:", len(df_pred_2025))
print("상장폐지 예측 수:", df_pred_2025['Ensemble_Pred'].sum().item())
print("상장폐지 예측 비율: {:.2f}%".format(100 * df_pred_2025['Ensemble_Pred'].m
```

```
   MLP_Prob  GRU_Prob  TCN_Prob  Ensemble_Prob  Ensemble_Pred
0  0.001820  0.390074  0.016511       0.136135              0
1  0.011858  0.503369  0.000675       0.171967              0
2  0.001566  0.453771  0.408206       0.287848              0
3  0.003770  0.456668  0.054410       0.171616              0
4  0.003866  0.456775  0.037736       0.166126              0
5  0.003545  0.456658  0.010960       0.157054              0
6  0.003922  0.456657  0.035093       0.165224              0
총 샘플 수: 7
상장폐지 예측 수: 0
상장폐지 예측 비율: 0.00%
```

- 추후 Symbol을 추가하면 해당 Symbol 기업의 2025년도 상장폐지 확률 예측 가능.
- 2020-2024년도 모두 data에 NA가 적은 기업 7개 모두 2025년도 상장폐지 확률 low